SPRUCE BUDWORM RADAR SEGMENTATION

Research Team: Dr. Jennie Pearce, Dr. Jean-Noel Candau, Dr. Elaine Ho-Tassoni, Eyad Gad

THE PROBLEM

Spruce budworm outbreaks damage forests through severe defoliation. Migrating moths can travel long distances and spread outbreaks to new areas.

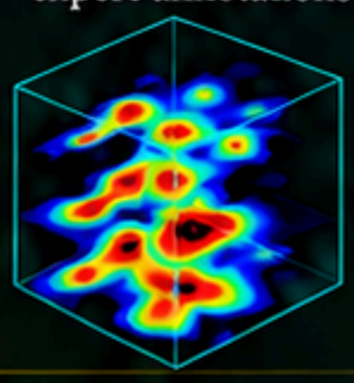
THE IDEA



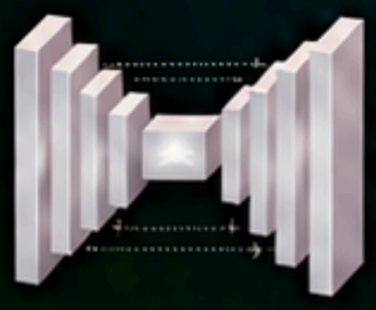
Use weather radar and deep segmentation to detect migrating budworm swarms and track their movement through the night.

METHOD WORKFLOW

1 OBSERVE
Weather radar volumes + expert annotations



2 TRAIN
Attention U-Net learns swarm patterns



3 PREDICT
Generate budworm segmentation mask



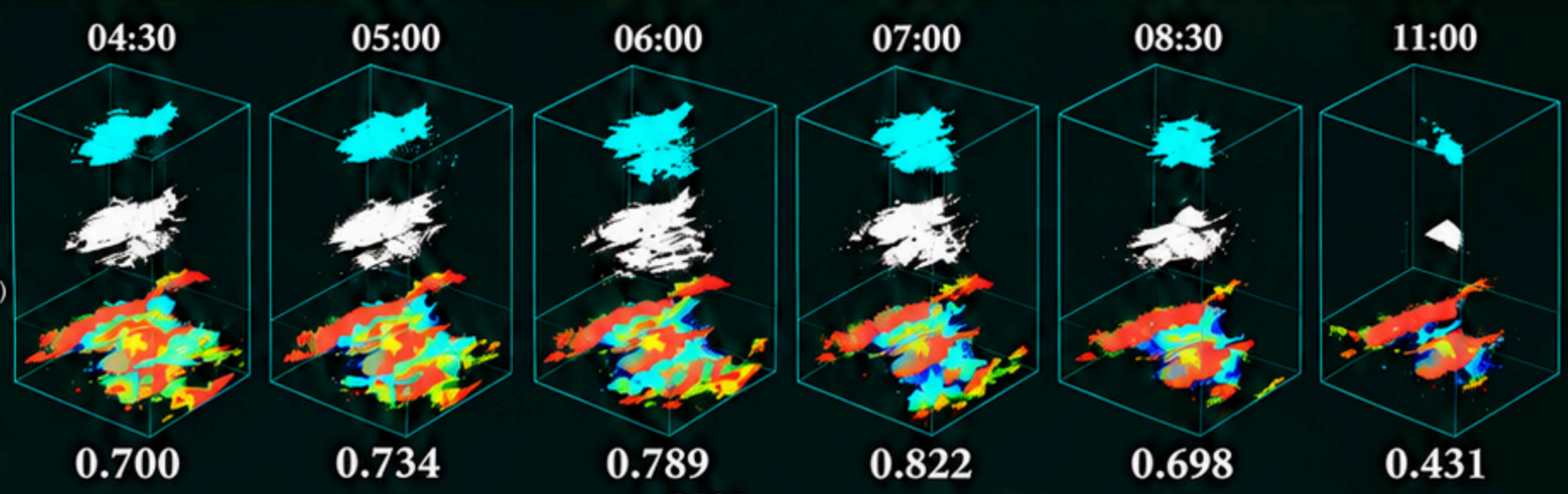
4 EVALUATE
Compare with expert labels using Dice, Precision, Recall



OVERALL RESULTS

2,052 RADAR SCENES	57 EXPERIMENTS
0.635 MACRO DICE	0.636 PRECISION
0.827 RECALL	

HOW A BUDWORM SWARM CHANGES THROUGH ONE NIGHT



Exact 480×480 pack values: no spatial smoothing. Vertical offests are display-only.

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